Lab 7 practice: Laplace transforms, operational properties and systems (Zill §4.2, 4.3, 4.6)

Prep for the Wed Oct 28 lab quiz. Closed book, about 10 minutes per problem. Lab 6 covered the basic pairs, partial fractions, derivatives, the s-shift and the unit step; here those steps are only the machinery. The new ideas are the derivative of a transform, convolution, periodic functions, the delta function and a 2 by 2 system. Every solution was checked with sympy (`lab7_check.py`).

From the course table (`reference/LaplaceTable.pdf`), with $F(s) = \mathcal{L}\{f(t)\}$:

$f(t)$	$\mathcal{L}\{f(t)\}$	$f(t)$	$\mathcal{L}\{f(t)\}$
1	$\dfrac{1}{s}$	$\dfrac{df}{dt}$	$sF(s) - f(0)$
t^n	$\dfrac{n!}{s^{n+1}}$	$\dfrac{d^2f}{dt^2}$	$s^2F(s) - sf(0) - f'(0)$
e^{at}	$\dfrac{1}{s-a}$	$e^{at}f(t)$	$F(s-a)$
$\cos kt$	$\dfrac{s}{s^2+k^2}$	$f(t-a)U(t-a)$	$e^{-as}F(s)$
$\sin kt$	$\dfrac{k}{s^2+k^2}$		

Not in the table, so know them. With $G(s) = \mathcal{L}\{g(t)\}$:

Property	Statement
derivative of a transform	$\mathcal{L}\{t^n f(t)\} = (-1)^n \dfrac{d^n F}{ds^n}$
convolution	$(f * g)(t) = \displaystyle\int_0^t f(\tau) g(t-\tau)\, d\tau$, and $\mathcal{L}\{f * g\} = F(s)G(s)$
integral	$\mathcal{L}\Big\{ \displaystyle\int_0^t f(\tau)\, d\tau \Big\} = \dfrac{F(s)}{s}$ (this is $1 * f$)
periodic, period T	$\mathcal{L}\{f\} = \dfrac{1}{1-e^{-sT}} \displaystyle\int_0^T e^{-st} f(t)\, dt$
Dirac delta, $a \geq 0$	$\mathcal{L}\{\delta(t-a)\} = e^{-as}$

System of two equations. Transform both, solve the 2 by 2 algebraic system for $X(s)$ and $Y(s)$ (Cramer's rule), then invert each.

Problem 1

Find (a) $\mathcal{L}\{t\sin 3t\}$, (b) $\mathcal{L}\{t^2\cos t\}$, (c) $\mathcal{L}\{te^{-t}\sin 3t\}$, and (d) $\mathcal{L}^{-1}\Big\{\dfrac{s}{(s^2+4)^2}\Big\}$.

Solution: derivative of the transform

(a) Pair $\sin kt$ with $k = 3$: $F(s) = \dfrac{3}{s^2+9}$. Multiplying by t means $-F'(s)$.

$$F'(s) = -\frac{6s}{(s^2+9)^2} \quad \Longrightarrow \quad \boxed{\mathcal{L}\{t\sin 3t\} = \frac{6s}{(s^2+9)^2}}$$

(b) Pair $\cos kt$ with $k = 1$: $F(s) = \dfrac{s}{s^2+1}$. Here $n = 2$, so the sign is $(-1)^2 = +1$ and we differentiate twice.

$$F'(s) = \frac{(s^2+1) - s(2s)}{(s^2+1)^2} = \frac{1-s^2}{(s^2+1)^2}$$

$$F''(s) = \frac{-2s}{(s^2+1)^2} + (1-s^2)\cdot\frac{-2\cdot 2s}{(s^2+1)^3} = \frac{-2s(s^2+1) - 4s(1-s^2)}{(s^2+1)^3} = \frac{2s^3-6s}{(s^2+1)^3}$$

$$\boxed{\mathcal{L}\{t^2\cos t\} = \frac{2s(s^2-3)}{(s^2+1)^3}}$$

(c) Two routes. Use the s-shift on the result of (a): $e^{-t}f(t)$ with $f = t\sin 3t$ gives $F(s+1)$, so replace s by $s+1$.

$$\mathcal{L}\{te^{-t}\sin 3t\} = \frac{6(s+1)}{\big((s+1)^2+9\big)^2}$$

Check by the other order: $\mathcal{L}\{e^{-t}\sin 3t\} = \dfrac{3}{(s+1)^2+9}$ (pair $e^{at}f(t)$), and $-\dfrac{d}{ds}$ of that is $\dfrac{3\cdot 2(s+1)}{((s+1)^2+9)^2}$, the same.

$$\boxed{\mathcal{L}\{te^{-t}\sin 3t\} = \frac{6(s+1)}{\big((s+1)^2+9\big)^2}}$$

(d) Run the property backwards. Pair $\sin kt$ gives $\mathcal{L}\{\frac{1}{2}\sin 2t\} = \dfrac{1}{s^2+4}$, and $-\dfrac{d}{ds}\dfrac{1}{s^2+4} = \dfrac{2s}{(s^2+4)^2}$. So $\dfrac{2s}{(s^2+4)^2} = \mathcal{L}\{t\cdot\frac{1}{2}\sin 2t\}$, and dividing by 2:

$$\boxed{\mathcal{L}^{-1}\Big\{\frac{s}{(s^2+4)^2}\Big\} = \frac{1}{4}\,t\sin 2t}$$

Problem 2

(a) Write $\mathcal{L}\Big\{\displaystyle\int_0^t e^{-\tau}\cos(t-\tau)\,d\tau\Big\}$ as a function of s without doing the integral. Then use the convolution theorem to find (b) $\mathcal{L}^{-1}\Big\{\dfrac{1}{(s^2+1)^2}\Big\}$ and (c) $\mathcal{L}^{-1}\Big\{\dfrac{1}{(s+1)(s^2+4)}\Big\}$.

Solution: convolution, forward and inverse

(a) The integral is $e^{-t} * \cos t$. Pairs e^{at} with $a = -1$ and $\cos kt$ with $k = 1$:

$$\boxed{\mathcal{L}\{e^{-t} * \cos t\} = \frac{1}{s+1} \cdot \frac{s}{s^2+1} = \frac{s}{(s+1)(s^2+1)}}$$

(b) Pair $\sin kt$ with $k = 1$: $\dfrac{1}{(s^2+1)^2} = \mathcal{L}\{\sin t\}\,\mathcal{L}\{\sin t\}$, so the inverse is $\sin t * \sin t$. Use $\sin A \sin B = \frac{1}{2}[\cos(A-B) - \cos(A+B)]$ with $A = \tau$, $B = t - \tau$:

$$\sin t * \sin t = \int_0^t \sin\tau\, \sin(t-\tau)\, d\tau = \frac{1}{2}\int_0^t \big[\cos(2\tau - t) - \cos t\big]\, d\tau$$

$$= \frac{1}{2}\Big[\tfrac{1}{2}\sin(2\tau - t) - \tau\cos t\Big]_0^t = \frac{1}{2}\Big[\tfrac{1}{2}\sin t + \tfrac{1}{2}\sin t - t\cos t\Big]$$

$$\boxed{\mathcal{L}^{-1}\Big\{\frac{1}{(s^2+1)^2}\Big\} = \frac{1}{2}(\sin t - t\cos t)}$$

(c) Pairs e^{at} with $a = -1$ and $\sin kt$ with $k = 2$: $\dfrac{1}{s^2+4} = \mathcal{L}\{\frac{1}{2}\sin 2t\}$. So

$$\mathcal{L}^{-1}\Big\{\frac{1}{(s+1)(s^2+4)}\Big\} = e^{-t} * \tfrac{1}{2}\sin 2t = \frac{1}{2}\int_0^t e^{-(t-\tau)}\sin 2\tau\, d\tau = \frac{1}{2}e^{-t}\int_0^t e^{\tau}\sin 2\tau\, d\tau$$

Integrating by parts twice, $\displaystyle\int e^{\tau}\sin 2\tau\, d\tau = \frac{e^{\tau}(\sin 2\tau - 2\cos 2\tau)}{5}$. Evaluating from 0 to t gives $\dfrac{e^{t}(\sin 2t - 2\cos 2t) + 2}{5}$, so

$$\boxed{\mathcal{L}^{-1}\Big\{\frac{1}{(s+1)(s^2+4)}\Big\} = \frac{1}{10}(\sin 2t - 2\cos 2t + 2e^{-t})}$$

Partial fractions give the same thing: the coefficient of $\dfrac{1}{s+1}$ is $\dfrac{1}{1+4} = \dfrac{1}{5}$, matching $\dfrac{2}{10}e^{-t}$. Convolution is the only sensible route for (b), where partial fractions over the reals fail because of the repeated factor $s^2 + 1$.

Problem 3

Solve (a) the integral equation $y(t) = t + \int_0^t \sin(t-\tau)\, y(\tau)\, d\tau$, and (b) the integrodifferential equation $y'(t) + 2y(t) + \int_0^t y(\tau)\, d\tau = t, \quad y(0) = 0.$

Solution: integral and integrodifferential equations

(a) The integral is a convolution, $\sin t * y$. Pair $\sin kt$ with $k = 1$, and $\mathcal{L}\{t\} = \dfrac{1}{s^2}$ from t^n. Transform both sides:

$$Y = \frac{1}{s^2} + \frac{1}{s^2+1}\,Y \quad\Longrightarrow\quad Y\,\frac{s^2}{s^2+1} = \frac{1}{s^2} \quad\Longrightarrow\quad Y = \frac{s^2+1}{s^4} = \frac{1}{s^2} + \frac{1}{s^4}$$

Pair t^n with $n = 1$ and $n = 3$ (note $\dfrac{1}{s^4} = \dfrac{1}{3!}\cdot\dfrac{3!}{s^4}$):

$$\boxed{y(t) = t + \frac{t^3}{6}}$$

(b) The integral of y is $1 * y$, with transform $\dfrac{Y}{s}$. Pair df/dt with $y(0) = 0$:

$$sY + 2Y + \frac{Y}{s} = \frac{1}{s^2} \quad\Longrightarrow\quad Y\,\frac{s^2+2s+1}{s} = \frac{1}{s^2} \quad\Longrightarrow\quad Y = \frac{1}{s(s+1)^2}$$

Partial fractions, $\dfrac{1}{s(s+1)^2} = \dfrac{A}{s} + \dfrac{B}{s+1} + \dfrac{C}{(s+1)^2}$: covering up gives $A = 1$ and $C = -1$, and $A + B = 0$ (no s^2 term in the numerator) gives $B = -1$.

$$Y = \frac{1}{s} - \frac{1}{s+1} - \frac{1}{(s+1)^2}$$

Pairs: 1, e^{at} with $a = -1$, and $e^{at}f(t)$ with $f = t$ for the last term.

$$\boxed{y(t) = 1 - e^{-t} - te^{-t}}$$

Check: $y(0) = 1 - 1 - 0 = 0$. Also $y' = te^{-t}$ and $\int_0^t y\,d\tau = t - 2 + 2e^{-t} + te^{-t}$, so

$$y' + 2y + \int_0^t y\,d\tau = te^{-t} + (2 - 2e^{-t} - 2te^{-t}) + (t - 2 + 2e^{-t} + te^{-t}) = t$$

Problem 4

Find the Laplace transform of each periodic function.

(a) The triangle wave with period 2: $f(t) = t$ for $0 \leq t < 1$, $f(t) = 2 - t$ for $1 \leq t < 2$, and $f(t+2) = f(t)$.

(b) The half-wave rectified sine: $f(t) = \sin t$ for $0 \leq t < \pi$, $f(t) = 0$ for $\pi \leq t < 2\pi$, and $f(t + 2\pi) = f(t)$.

Solution: periodic functions

(a) $T = 2$. Antiderivatives:

$$\int te^{-st}\,dt = -\Big(\frac{t}{s} + \frac{1}{s^2}\Big)e^{-st}, \qquad \int (2-t)e^{-st}\,dt = -\Big(\frac{2-t}{s} - \frac{1}{s^2}\Big)e^{-st}$$

$$\int_0^1 te^{-st}\,dt = \frac{1}{s^2} - \Big(\frac{1}{s} + \frac{1}{s^2}\Big)e^{-s}$$

$$\int_1^2 (2-t)e^{-st}\,dt = \frac{e^{-2s}}{s^2} + \Big(\frac{1}{s} - \frac{1}{s^2}\Big)e^{-s}$$

The $\dfrac{e^{-s}}{s}$ terms cancel in the sum:

$$\int_0^2 e^{-st} f\,dt = \frac{1 - 2e^{-s} + e^{-2s}}{s^2} = \frac{(1-e^{-s})^2}{s^2}$$

Divide by $1 - e^{-2s} = (1-e^{-s})(1+e^{-s})$:

$$\boxed{F(s) = \frac{1}{s^2} \cdot \frac{1-e^{-s}}{1+e^{-s}} = \frac{\tanh(s/2)}{s^2}}$$

(b) $T = 2\pi$, and $f = 0$ on the second half, so only $[0, \pi]$ contributes. The antiderivative is

$$\int e^{-st}\sin t\,dt = \frac{e^{-st}(-s\sin t - \cos t)}{s^2+1}$$

so

$$\int_0^\pi e^{-st}\sin t\,dt = \frac{e^{-s\pi}(0+1) - (0-1)}{s^2+1} = \frac{1+e^{-\pi s}}{s^2+1}$$

$$F(s) = \frac{1+e^{-\pi s}}{(s^2+1)(1-e^{-2\pi s})}$$

Since $1 - e^{-2\pi s} = (1-e^{-\pi s})(1+e^{-\pi s})$, the factor $1 + e^{-\pi s}$ cancels:

$$\boxed{F(s) = \frac{1}{(s^2+1)(1-e^{-\pi s})}}$$

Cancel the common factor before stopping. The unsimplified form is correct but the short one is what a marker looks for.

Problem 5

Solve $y'' + 4y = \sin t + \delta(t - \pi), \quad y(0) = 1, \quad y'(0) = 0$, and state the jump in y' at $t = \pi$.

Solution: impulse forcing

The delta is a unit impulse at $t = \pi$. Transform with pair d^2f/dt^2, pair $\sin kt$ with $k = 1$, and $\mathcal{L}\{\delta(t-\pi)\} = e^{-\pi s}$:

$$s^2Y - s\cdot 1 - 0 + 4Y = \frac{1}{s^2+1} + e^{-\pi s} \quad\Longrightarrow\quad Y = \frac{s}{s^2+4} + \frac{1}{(s^2+1)(s^2+4)} + \frac{e^{-\pi s}}{s^2+4}$$

Partial fractions for the middle term: $\dfrac{1}{(s^2+1)(s^2+4)} = \dfrac{1}{3}\Big(\dfrac{1}{s^2+1} - \dfrac{1}{s^2+4}\Big)$. Invert with pairs $\cos kt$ ($k=2$) and $\sin kt$ ($k=1$ and $k=2$, using $\dfrac{1}{s^2+4} = \mathcal{L}\{\frac{1}{2}\sin 2t\}$). For the last term use $f(t-a)U(t-a)$ with $a=\pi$ and $f = \frac{1}{2}\sin 2t$; since $\sin 2(t-\pi) = \sin 2t$, the shifted term is $\frac{1}{2}\sin 2t\,U(t-\pi)$.

$$y(t) = \cos 2t + \frac{1}{3}\sin t - \frac{1}{6}\sin 2t + \frac{1}{2}\sin 2t\;U(t-\pi)$$

$$\boxed{y(t) = \begin{cases} \cos 2t + \frac{1}{3}\sin t - \frac{1}{6}\sin 2t, & 0 \le t < \pi \\ \cos 2t + \frac{1}{3}\sin t + \frac{1}{3}\sin 2t, & t \ge \pi \end{cases}}$$

Jump. y itself is continuous at π (the added term is $\frac{1}{2}\sin 2t$, which is 0 at $t=\pi$). Its derivative gains $\frac{1}{2}\cdot 2\cos 2\pi = 1$, so $y'(\pi^+) - y'(\pi^-) = 1$. That is the rule for $\delta(t-a)$ in $y'' + ky = \ldots$: the velocity jumps by the strength of the impulse and the position does not.

Problem 6

Solve the system with the Laplace transform:

$$x' = 3x - 2y, \qquad y' = 2x - 2y + 1, \qquad x(0) = 1, \quad y(0) = 0$$

Solution: 2 by 2 system

Transform each equation with pair df/dt and $\mathcal{L}\{1\} = \dfrac{1}{s}$:

$$sX - 1 = 3X - 2Y, \qquad sY - 0 = 2X - 2Y + \frac{1}{s}$$

Collect the unknowns on the left:

$$(s-3)X + 2Y = 1, \qquad -2X + (s+2)Y = \frac{1}{s}$$

The determinant is $\Delta = (s-3)(s+2) + 4 = s^2 - s - 2 = (s-2)(s+1)$. Cramer's rule:

$$X = \frac{1 \cdot (s+2) - 2 \cdot \frac{1}{s}}{\Delta} = \frac{s^2 + 2s - 2}{s(s-2)(s+1)}, \qquad Y = \frac{(s-3) \cdot \frac{1}{s} - (-2)(1)}{\Delta} = \frac{3(s-1)}{s(s-2)(s+1)}$$

Partial fractions by covering up each factor. For X:

$$X = \frac{1}{s} + \frac{1}{s-2} - \frac{1}{s+1}$$

at $s = 0$: $\dfrac{-2}{(-2)(1)} = 1$; at $s = 2$: $\dfrac{6}{2 \cdot 3} = 1$; at $s = -1$: $\dfrac{-3}{(-1)(-3)} = -1$. For Y: at $s = 0$: $\dfrac{-3}{(-2)(1)} = \dfrac{3}{2}$; at $s = 2$: $\dfrac{3}{2 \cdot 3} = \dfrac{1}{2}$; at $s = -1$: $\dfrac{-6}{(-1)(-3)} = -2$.

$$Y = \frac{3/2}{s} + \frac{1/2}{s-2} - \frac{2}{s+1}$$

Invert with pairs 1 and e^{at}:

$$\boxed{x(t) = 1 + e^{2t} - e^{-t}, \qquad y(t) = \frac{3}{2} + \frac{1}{2}e^{2t} - 2e^{-t}}$$

Check: $x(0) = 1$, $y(0) = 0$. Also $x' = 2e^{2t} + e^{-t}$, and $3x - 2y = 3 + 3e^{2t} - 3e^{-t} - 3 - e^{2t} + 4e^{-t} = 2e^{2t} + e^{-t}$. The second equation is the same kind of check, with $y' = e^{2t} + 2e^{-t}$.

Problem 7

Solve $ty'' - ty' + y = 2, \quad y(0) = 2, \quad y'(0) = -4.$

Solution: variable coefficients, transform ODE is first order

The coefficients depend on t, so each $t\,(\cdot)$ term becomes $-\dfrac{d}{ds}$ of a transform. With $\mathcal{L}\{y\} = Y$:

$$\mathcal{L}\{y''\} = s^2Y - 2s + 4, \qquad \mathcal{L}\{y'\} = sY - 2$$

$$\mathcal{L}\{ty''\} = -\frac{d}{ds}(s^2Y - 2s + 4) = -s^2Y' - 2sY + 2$$

$$\mathcal{L}\{-ty'\} = +\frac{d}{ds}(sY - 2) = Y + sY'$$

The constant $y'(0) = -4$ disappears under $\dfrac{d}{ds}$, so it does not enter this equation. Add everything, with $\mathcal{L}\{y\} = Y$ and $\mathcal{L}\{2\} = \dfrac{2}{s}$:

$$(s - s^2)Y' + (2 - 2s)Y + 2 = \frac{2}{s} \quad \Longrightarrow \quad (1 - s)(sY' + 2Y) = \frac{2(1 - s)}{s}$$

Cancel $1 - s$ (valid for $s \neq 1$): $sY' + 2Y = \dfrac{2}{s}$. Multiply by s to make the left side an exact derivative:

$$s^2Y' + 2sY = (s^2Y)' = 2 \quad \Longrightarrow \quad s^2Y = 2s + C \quad \Longrightarrow \quad Y = \frac{2}{s} + \frac{C}{s^2}$$

Pairs 1 and t^n with $n = 1$: $y = 2 + Ct$. Now $y'(0) = C = -4$ fixes the constant, and $y(0) = 2$ is already satisfied.

$$\boxed{y(t) = 2 - 4t}$$

Check: $y'' = 0$, $y' = -4$, so $ty'' - ty' + y = 0 + 4t + 2 - 4t = 2$.

Problem 8

Solve $y'' + 4y = t \sin 2t, \quad y(0) = 0, \quad y'(0) = 0.$

Solution: the forcing needs the derivative rule, the inverse needs convolution

The forcing is resonant (frequency 2 on a system with natural frequency 2). From problem 1 with $k = 2$, $\mathcal{L}\{t \sin 2t\} = -\dfrac{d}{ds}\dfrac{2}{s^2+4} = \dfrac{4s}{(s^2+4)^2}$. With zero initial data:

$$(s^2+4)Y = \frac{4s}{(s^2+4)^2} \quad \Longrightarrow \quad Y = \frac{4s}{(s^2+4)^3} = -\frac{d}{ds}\left[\frac{1}{(s^2+4)^2}\right]$$

Let $G(s) = \dfrac{1}{(s^2+4)^2}$ with $g = \mathcal{L}^{-1}\{G\}$. Then $Y = -G'$, so $y = t\,g(t)$. Find g by convolution: $G = \left(\dfrac{1}{s^2+4}\right)^2 = \frac{1}{4}\,\mathcal{L}\{\sin 2t\}^2$ (pair $\sin kt$ with $k = 2$), so $g = \frac{1}{4}\,\sin 2t * \sin 2t$.

$$\sin 2t * \sin 2t = \frac{1}{2}\int_0^t \left[\cos(4\tau - 2t) - \cos 2t\right] d\tau = \frac{1}{2}\left[\tfrac{1}{4}\sin(4\tau - 2t) - \tau\cos 2t\right]_0^t = \frac{1}{4}\sin 2t - \frac{t}{2}\cos 2t$$

so $g(t) = \dfrac{1}{16}(\sin 2t - 2t\cos 2t)$ and

$$\boxed{y(t) = t\,g(t) = \frac{t\sin 2t}{16} - \frac{t^2\cos 2t}{8}}$$

Check: $y(0) = 0$, and $y' = \dfrac{\sin 2t + 2t\cos 2t}{16} - \dfrac{2t\cos 2t - 2t^2\sin 2t}{8}$ is 0 at $t = 0$. The $t^2\cos 2t$ term is the resonance: the amplitude grows without bound.